

Function of 1 variable: $f(x) = \sin(x)$

Example: $* f(x,y) = x^2 + y^2$

* $f(x, y) = \sqrt{y}$ only defined if $y \geq 0$

* $f(x,y) = \frac{1}{x+y}$ only defined if $x+y \neq 0$

* $f(x,y)$ = temperature at point (x,y)

How to visualise f with f a function of 2 variables?

graph: $z = f(x, y)$



Example: $f(x,y) = -y$ graph: $z = -y$

Example 2: $f(x,y) = 1 - x^2 - y^2$: in y plane: $x=0$ in x plane: $y=0$
 $z = 1 - y^2$ $z = 1 - x^2$

When does it hit xy plane? $z=0$

$$1 - x^2 - y^2 = 0$$
$$\Leftrightarrow \underbrace{x^2 + y^2 = 1}_{\text{Unit circle}}$$

Contours plot:



Shows all the points where $F(x,y)$: some fixed constraints chosen at regular intervals

We slice the graph by horizontal planes $z=c$, level curves of f

Example: $f(x, y) = -y$

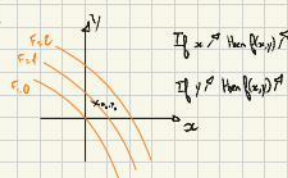
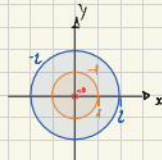


$$f(x,y) = 1 - (x^2 + y^2)$$

$$f=0 \Leftrightarrow x^2+y^2=1$$

$$\begin{cases} = 1 \Leftrightarrow x^2 + y^2 = 0 \\ = -1 \Leftrightarrow x^2 + y^2 = 1 \end{cases}$$

$$f = (-1)(-2) \quad x^2 + y^2 = 2$$



If $x \in \text{Horn}(f(x, y))$

$\exists y \in \text{Kern } f(x, y) \neq$

Partial derivatives: function of 1 variable: $f'(x) = \frac{df}{dx} \Leftrightarrow \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$



Approximation formula $z \approx f(z_0)$

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0)$$

Question: Do the same for $f(x,y)$?

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

$$\frac{\partial f}{\partial y}(x_0, y_0) = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y}$$